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Mayank Raj

Ph.D. Research Summary

My PhD research lies at the intersection of machine learning, applied mathematics, and material science. I develop data-driven and physics-informed machine-learning methods for discovering models of complex materials using computational and experimental data. The particular focus of my research is on learning governing laws, constitutive relations, and latent variables while incorporating physical structure and mathematical constraints. Find my Ph.D candidacy slides here and my projects portfolio here.

Education

California Institute of Technology — Pasadena, CA, USA
*Ph.D. in **Applied and Computational Mathematics*** — 2022 – 2027
- Advisor: Prof. Andrew M. Stuart
- GPA: 4.1

Indian Institute of Technology Madras — Chennai, India
*M.Tech. in **Data Science*** — 2020 – 2021
*B.Tech. in **Mechanical Engineering*** — 2016 – 2020
- Minor in **Applied Mathematics**
- GPA: 9.28/10

Employment History

Wells Fargo — Bangalore, India
Data Scientist — 2021 – 2022
- Developed an **automated OCR and regular-expression pipeline** to detect, extract, and structure data from unstructured PDF documents.
- Deployed the pipeline on the firm's **parallel computing** cluster, automating the end-to-end workflow from PDF ingestion and data extraction to structured-data generation.

Internships

Tata Research Design and Development Center — Pune, India
Research Intern in Machine Learning, Project Website — Winter, 2020
- Developed a **machine-learning-driven framework for multi-objective optimization** of product designs, integrating CFD simulations with surrogate modeling.
- Built a **machine learning surrogate** model to rapidly predict separation factor and pressure drop, enabling efficient exploration of the particle-classifier design space.
- Performed **multi-objective optimization** to identify Pareto-optimal designs balancing separation performance against pressure drop.

RWTH Aachen University — Aachen, Germany
Research Intern in Computational Mechanics, Project Website — Summer, 2019
- Modeled the 3D brick-and-mortar microstructure of nacre using HyperMesh, representing the composite architecture of brittle calcium carbonate and ductile organic matrix.
- Conducted LS-DYNA simulations to characterize the mechanical properties and investigate the effect of material composition on the toughness of nacre-based composites.
- Demonstrated that increasing the organic-matrix fraction enhances composite toughness but results in a corresponding reduction in overall strength.

Publications

A full list of publications on Google Scholar.

[5] **Mayank Raj** et al. "A neural-network framework to learn history-dependent constitutive laws and identifiability of internal variables". *Journal of the Mechanics and Physics of Solids* 217 (2026), p. 106807. ISSN: 0022-5096. DOI: https://doi.org/10.1016/j.jmps.2026.106807.

[4] Harkirat Singh, **Mayank Raj**, and Kaushik Bhattacharya. "Effective behavior of heterogeneous media governed by strain gradient elasticity". *Journal of the Mechanics and Physics of Solids* 212 (2026), p. 106583. ISSN: 0022-5096. DOI: https://doi.org/10.1016/j.jmps.2026.106583.

[3] **Mayank Raj**, Pramod Kumbhar, and Ratna Kumar Annabattula. "Physics-informed neural networks for solving thermo-mechanics problems of functionally graded material" (2022). arXiv: 2111.10751 [cs.CE].

[2] **Mayank Raj** et al. "Estimation of Local Strain Fields in Two-Phase Elastic Composite Materials Using UNet-Based Deep Learning". *Integrating Materials and Manufacturing Innovation* 10.3 (Sept. 1, 2021), pp. 444–460. DOI: 10.1007/s40192-021-00227-2.

[1] **Mayank Raj**, Sandeep P. Patil, and Bernd Markert. "Mechanical Properties of Nacre-Like Composites: A Bottom-Up Approach". *Journal of Composites Science* 4.2 (2020). ISSN: 2504-477X. DOI: 10.3390/jcs4020035.

Awards & Achievements

- ❑ The Bell Family Fellowship — 2022
- ❑ Mitacs Globalink Scholarship — 2020
- ❑ UGC-DAAD Research Scholarship — 2019
- ❑ Top 0.15% in JEE-Advanced — 2016

Teaching Assistant

Linear Analysis with Applications; Signal-Processing Systems and Transforms; Introductory Methods of Applied Mathematics for the Physical Sciences; Introduction to Computational Science & Engineering

Relevant Coursework

Probability, Statistics & Stochastic Processes; Stochastic Differential Equations; Linear Algebra; Optimization; Deep Learning

Conferences, Workshops & Summer Schools

- ❑ Frontiers in Applied Analysis, Carnegie Mellon University — 2025
- ❑ Machine Learning Theory Summer School, Princeton University — 2025

Tools

PyTorch, Numpy, Pandas, Scipy, Scikit Learn

References

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